



LAL
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Lecture-2

“VB. Net”

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Advanced Technology (Dot Net) Lab

1. Familiarization with IDE.
2. Programming Console applications using VB.NET covering all the aspects

of VB.NET fundamentals

3. Object oriented programming using VB.NET covering objects, Inheritance, Polymorphism,

Constructors, Static Classes, and Interfaces.

4. Programme to illustrate Exception Handling concepts.

5. Programme to illustrate use of Collections.

6. Programme to perform File I/O Operations.

7. Programming Windows applications using VB.NET covering all major controls and components,

Menus, MDI, Event Handling.

8. Creating windows installer.

9. Programme to interact with Database from a Windows Desktop Application.

10. Programming to build web applications using web controls, maintaining state.

Note: The Instructor may add/delete/modify/tune experiments, wherever he/she feels in a justified manner.

What is VB.NET?

- The VB.NET stands for **Visual Basic. Network Enabled Technologies.**
- It is a simple, high-level, **object-oriented programming** language developed by Microsoft in 2002.
- Furthermore, it supports the OOPs concept, such as abstraction, encapsulation, inheritance, and polymorphism. Therefore, everything in the VB.NET language is an object, including all primitive data types (Integer, String, char, long, short, Boolean, etc.), user-defined data types, events, and all objects that inherit from its base class.
- It is not a case sensitive language, whereas, C++, Java, and C# are case sensitive language.

VB.NET Hello World Program

- A VB.NET define the following structure to create a program:
- Namespace declaration
- Procedure can be multiple
- Define a class or module
- Variables
- The Main procedure
- Statement and Expression
- Comments
- Create a **Hello_Program.vb** file in MYConsoleApp1 project and write the following code:

Hello_Program.vb

```
Imports System 'System is a Namespace
Module Hello_Program

    Sub Main()

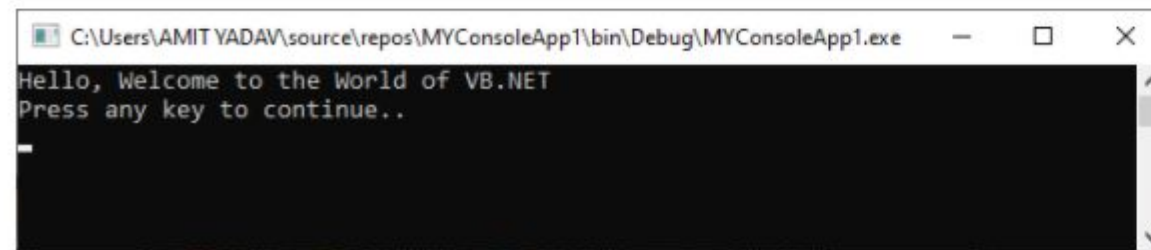
        Console.WriteLine("Hello, Welcome to the world of VB.NET")
        Console.WriteLine("Press any key to continue...")
        Console.ReadKey()

    End Sub

End Module
```

Let's compile and run the above program by pressing the F5 key, we get the following output.

Output:

A screenshot of a Windows console window. The title bar shows the file path: C:\Users\AMIT YADAV\source\repos\MYConsoleApp1\bin\Debug\MYConsoleApp1.exe. The console output consists of two lines: "Hello, Welcome to the World of VB.NET" and "Press any key to continue..". A cursor is visible on the line following the second line of output.

```
C:\Users\AMIT YADAV\source\repos\MYConsoleApp1\bin\Debug\MYConsoleApp1.exe
Hello, Welcome to the World of VB.NET
Press any key to continue..
_
```

- In VB.NET programming, the first line of the program is "Import System", where Imports is a statement that inherit the system namespace. A System is a namespace that contains basic classes, reference data types, events, attributes, and various inbuilt functions that help to run the program.
 - The Second line defines the Module It specifies the VB.NET filename. As we know, VB.NET language is a completely object-oriented language, so every program must contain a module or class in which you can write your program that contains data and procedures within the module.
-

Module Module1

End Module

- You can define more than one procedure in classes and modules. Generally, the procedure contains executable code to run. A procedure may contain the following function:

-
- Function
 - Operator
 - Sub
 - Get
 - Set
 - AddHandler
 - RemoveHandler

- Every program must contain a Main() method. In VB.NET, there is a Main() method or procedure that represents the starting point to execute a program, as we have seen in C language, their entry point is the main() function.
- A comment (') symbol is used to comment on a line that is ignored by the compiler in a program, and it is a good practice to use comments for a better understanding of the program.
- The Console.WriteLine() is a method of the console class. It is used to print any text or messages in the application. And the Console.ReadKey() is used to read a single character from the keyboard back to the Visual Studio IDE.

VB. Net is an object-oriented programming language

- **Object** – Objects have states and behaviors. **Example: A dog has states - color, name, breed as well as behaviors - wagging, barking, eating, etc.** An object is an instance of a class.
- **Class** – A class can be defined as a template/blueprint that describes the behaviors/states that objects of its type support.
- **Methods** – A method is basically a behavior. A class can contain many methods. It is in methods where the logics are written, data is manipulated and all the actions are executed.
- **Instance Variables** – Each object has its unique set of instance variables. An object's state is created by the values assigned to these instance variables.

Elements

- Identifiers
- Keywords

Data Types in VB. Net

Boolean	Depends on implementing platform	True or False
Byte	1 byte	0 through 255 (unsigned)
Char	2 bytes	0 through 65535 (unsigned)
Date	8 bytes	0:00:00 (midnight) on January 1, 0001 through 11:59:59 PM on December 31, 9999
Decimal	16 bytes	0 through +/-79,228,162,514,264,337,593,543,950,335 (+/-7.9...E+28) with no decimal point; 0 through +/-7.9228162514264337593543950335 with 28 places to the right of the decimal
Double	8 bytes	-1.79769313486231570E+308 through -4.94065645841246544E-324, for negative values 4.94065645841246544E-324 through 1.79769313486231570E+308, for positive values
Integer	4 bytes	-2,147,483,648 through 2,147,483,647 (signed)
Long	8 bytes	-9,223,372,036,854,775,808 through 9,223,372,036,854,775,807(signed)

Variables

- The Dim statement is used for variable declaration and storage allocation for one or more variables. The Dim statement is used at module, class, structure, procedure or block level.
- **Syntax for variable declaration in VB.Net is –**
- [< attributelist >] [accessmodifier] [[Shared] [Shadows] | [Static]]
- [ReadOnly] Dim [WithEvents] variablelist

Some valid variable declarations along with their definition are shown here –

- Dim StudentID As Integer
- Dim StudentName As String
- Dim Salary As Double
- Dim count1, count2 As Integer
- Dim status As Boolean
- Dim exitButton As New System.Windows.Forms.Button
- Dim lastTime, nextTime As Date

Thank You